

Machine-Learning AES Key Recovery by Side-Channel Analysis on the ASCAD Dataset

A replication of Poudel & Rahimi (arXiv:2508.11817, 2025)

[Name] | [Roll no.] | [Course / code] | September 2026

Abstract

This project reproduces the profiled side-channel attack of Poudel and Rahimi on the ASCAD dataset, in which one byte of an AES-128 key is recovered from electromagnetic power traces of a masked software implementation. The task is posed as 256-class classification of the first-round S-box output $Sbox(p \oplus k)$, and success is measured by guessing entropy (key rank) rather than classification accuracy. We built the full pipeline from the raw HDF5 files, trained four model families (random forest, SVM, a compact CNN and a 1-D ResNet), and reproduced the paper's central qualitative claims: the deep models recover the key byte while the classical baselines do not, and the ResNet is more trace-efficient than the CNN on the harder variable-key set. Beyond the paper we added a multi-task ResNet, a desynchronisation robustness test, and a shuffled-label control. The multi-task network cuts the traces needed on the fixed-key set from about 870 to about 270, but does not help on the variable-key set. All code, configurations and result files accompany this report.

1 Introduction

A masking countermeasure splits every sensitive intermediate value into random shares so that no single point in a power trace depends on the secret. ASCAD (ANSSI, 2018) is the reference benchmark for evaluating whether a profiled, learning-based attacker can still recover the key from such an implementation. Poudel and Rahimi frame the attack as supervised classification of the S-box output for one key byte and report key recovery within a few tens of attack traces on ASCAD.

The goal here was to rebuild that attack end to end and check which of its claims hold when the pipeline is written from scratch. The work was organised as three milestones: (1) classical baselines plus a CNN on the fixed-key set, (2) a 1-D ResNet and the variable-key set, and (3) a set of extensions - an SVM baseline, a multi-byte assembly step, a multi-task ResNet, a desynchronisation test, and sanity controls.

2 Background

2.1 Target and leakage model

The attacked value is $y = Sbox(p[b] \oplus k[b])$ for a chosen key byte b , where p is the known plaintext. Under the identity leakage model this gives 256 classes. Because ASCAD only masks bytes 2 to 15, and byte 2 is the community-standard target, all experiments use $b = 2$ unless stated otherwise. The stored labels in ASCAD.h5 are for byte 2 under this model; we re-derive labels from metadata and cross-check them against the stored ones.

2.2 Why not accuracy

On masked ASCAD the top-1 accuracy of any model sits near the $1/256 = 0.39\%$ chance line, so accuracy is not a useful metric. The standard measure is guessing entropy: for a set of attack traces and every key hypothesis g in $[0, 255]$, the log-probabilities the model assigns to $Sbox(p_i \oplus g)$ are summed, the 256 hypotheses are ranked by that score, and the rank of the true key byte is recorded. Rank 0 means the key byte is recovered. The curve of mean rank against number of attack traces, averaged over many random trace orderings, is the primary result.

2.3 Model selection

Validation cross-entropy is a poor stopping criterion here: for roughly the first 20 epochs it is flat or rising while the attack is already improving, so the lowest-loss epoch corresponds to a near-untrained network. Every deep model in this project is instead checkpointed on validation guessing entropy computed on a held-out set with the same key, evaluated every few epochs.

3 Method

The pipeline reads the ASCAD HDF5 files into profiling and attack sets, standardises the traces using statistics fitted on the profiling set only, and builds labels for the chosen byte. Four model families share one interface:

Random forest (300 trees) and **SVM** (RBF kernel, PCA to 50 components, profiling subsampled to 10 000 traces because an RBF SVM is quadratic in sample count) are the classical baselines. The **CNN** is the compact ASCAD network of Zaid et al. (about 17k parameters, SELU, one-cycle learning rate); the larger VGG-style ASCAD_best network was tried but memorises the profiling set without ranking the key. The **ResNet** is a 1-D residual network in the style of Karayalçin et al.: kernel size 11, SELU with LeCun-normal initialisation, a number of residual blocks scaled as $\text{floor}(\log_2(\text{length})) - 2$, a flatten head, and a triangular cyclic learning rate. Training keeps the trace tensor resident on the GPU, which is several times faster for these small models on large trace sets.

Key rank is computed by `src/key_rank.py` and every run writes a resolved configuration, a log, the fitted model, a training-history plot, the key-rank curve as CSV and PNG, and a metrics file.

4 Experimental setup

Fixed-key ASCAD (ATMega8515) provides 50 000 profiling and 10 000 attack traces of 700 samples; the variable-key set (ASCAD-r) provides 200 000 and 100 000 traces of 1400 samples. Desynchronised variants shift each trace randomly by up to 50 or 100 samples. Training ran on an NVIDIA RTX 4060 (8 GB) with PyTorch 2.14 and CUDA 12.6, inside a project-local virtual environment. Key rank is averaged over 100 random attack-trace orderings.

5 Results

5.1 Milestone 1 - baselines and CNN (fixed key)

On the fixed-key set the random forest never converges: its mean key rank is still about 63 after 2000 traces. The SVM baseline added in milestone 3 is no better, ending at a mean rank of 138 (chance is 127.5). The compact CNN recovers the key byte, reaching rank 0 at roughly 1360 attack traces, with the 10-90 percentile band at zero by about 700. This matches the paper's qualitative split between classical and deep models.

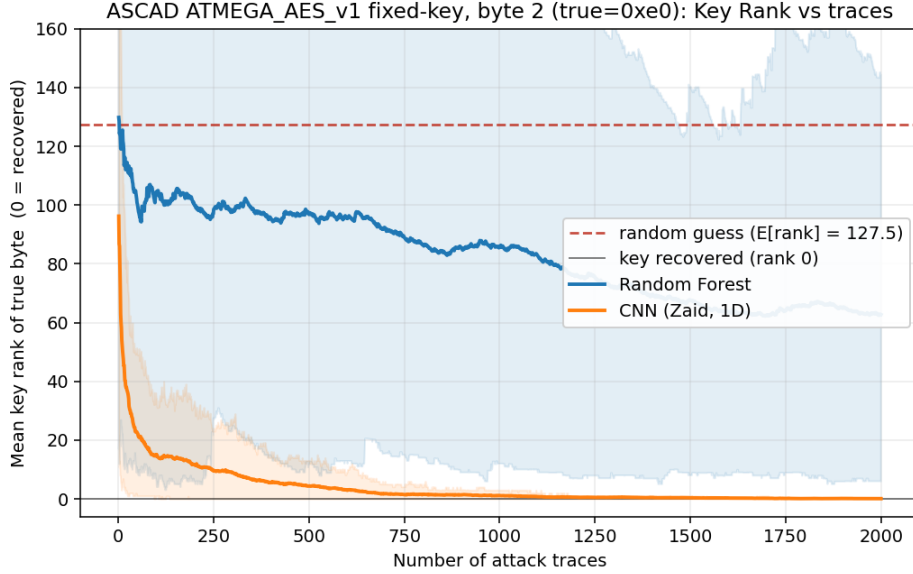


Figure 1. Fixed-key ASCAD, byte 2. The random forest stays near chance; the compact CNN drives the key rank to 0.

5.2 Milestone 2 - ResNet and the variable-key set

The rebuilt 1-D ResNet recovers the key on both datasets. On the fixed-key set it reaches rank 0 at about 870 traces; on the harder variable-key set it needs about 280, against about 520 for the CNN. That the ResNet is more trace-efficient on the harder problem is the paper's main qualitative point, and it is reproduced here, though our trace counts are roughly an order of magnitude above the best figures in the literature (around 35), so there is clear tuning headroom. The variable-key ResNet figure is also seed-sensitive: a second training seed needed about 1540 traces rather than 280, so the lower number should be read as a lucky run rather than a stable result. An earlier kernel-3 / ReLU ResNet overfit the fixed-key set to chance and would not train on the variable-key set at all; the fixes that made it work (larger kernels, SELU, depth scaled to the input length, cyclic learning rate, and judging convergence by validation guessing entropy rather than the first few epochs) are documented in the repository.

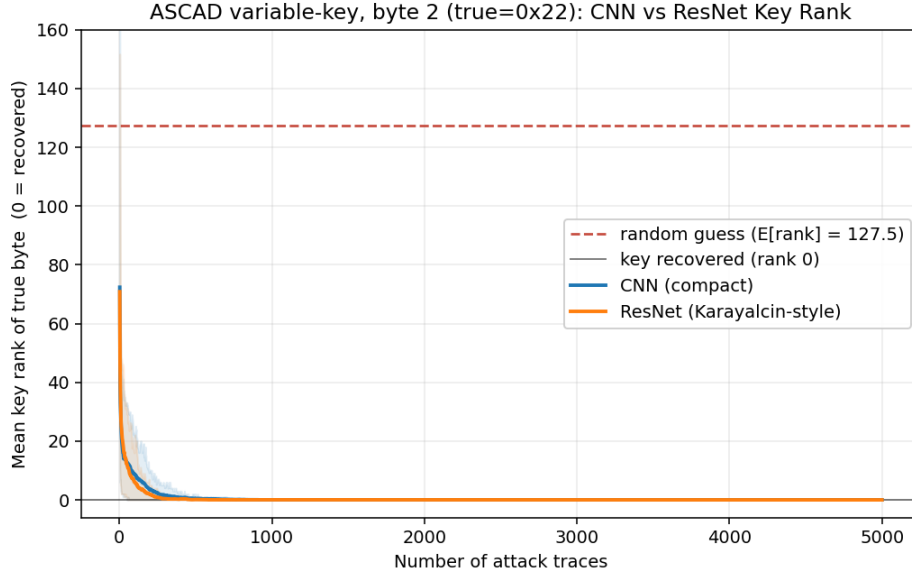


Figure 2. Variable-key ASCAD, byte 2. The ResNet recovers the key byte in fewer traces than the CNN.

5.3 Milestone 3a - multi-task ResNet

The masking scheme exposes two shares at the targeted sample: the mask r_{out} and the masked S-box output $y \oplus r_{\text{out}}$. The multi-task ResNet keeps one shared convolutional trunk and attaches three linear heads that are trained jointly - one for y , one for r_{out} , and one for $y \oplus r_{\text{out}}$ - but only the y head is used for key recovery. Predicting a raw mask byte is easy, so this head gives back-propagation a non-flat gradient during the early plateau. On the fixed-key set the effect is large: validation guessing entropy reaches 0 by epoch 10 instead of about 25, and the key byte is recovered in about 270 traces rather than 870. On the variable-key set the plateau is not the bottleneck and the extra heads act as noise; the single-task ResNet stays well ahead. This is reported as a negative result for that setting.

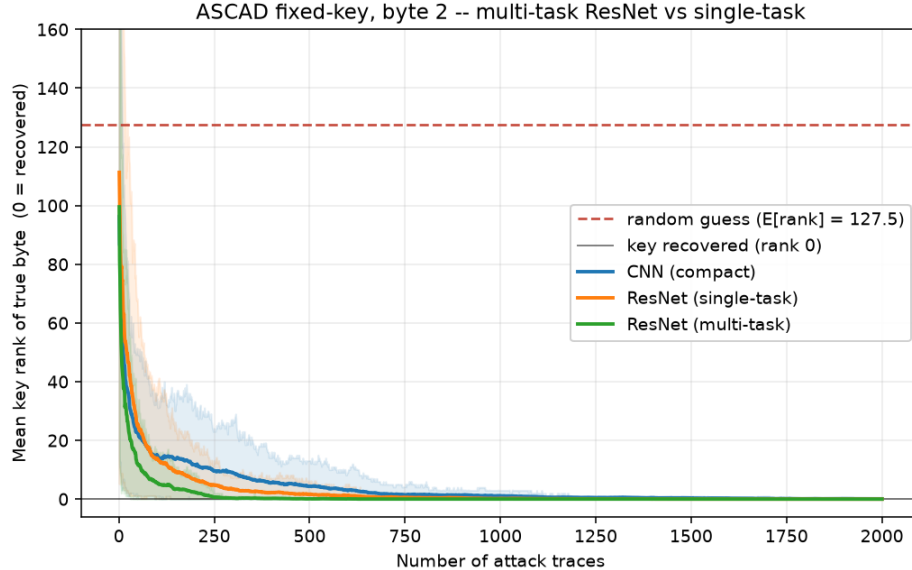


Figure 3. Fixed-key ASCAD, byte 2. The multi-task ResNet reaches rank 0 roughly three times sooner than the single-task ResNet.

5.4 Milestone 3b - how many key bytes can be recovered

The published ASCAD databases are a short trace window around the masked S-box operation of key byte 2. A CNN trained on byte 3 (mean rank about 170) or byte 4 (about 160) never leaves chance, on both the fixed-key and variable-key sets, because the leakage for those bytes falls outside the window. Recovering the

full 16-byte key would need the roughly 60 GB raw trace set and a per-byte re-windowing step, which was out of scope. The assembly step is still implemented end to end: `src/assemble_key.py` runs one model per byte, takes the rank-0 hypothesis as the recovered byte, stitches the 16 bytes together and checks them against the key in the metadata.

For the fixed-key set the recovered byte 2 matches the ground truth:

```
true key   : 4d fb e0 f2 72 21 fe 10 a7 8d 4a dc 8e 49 04 69
recovered  : -- -- e0 -- -- -- -- -- -- -- -- -- -- -- --
```

5.5 Milestone 3c - desynchronisation

When the attack traces are randomly shifted the compact CNN collapses: its mean key rank ends at 109 for a shift of up to 50 samples and 176 for up to 100 samples, with attack accuracy back at the 1/256 chance level, and the key is never recovered. The ResNet still recovers it - rank 0 after about 608 traces at shift 50 and about 979 at shift 100, against about 870 on aligned traces, so a modest penalty rather than a failure. This matches the literature: a deeper residual network tolerates misalignment that a compact CNN cannot. The caveat is training stability - the shift-50 ResNet run only recovered on the second attempt, after its guessing-entropy plateau failed to break inside a 50-epoch budget and did break at 80 epochs. Shift augmentation during training would be the next step to close the remaining gap to the aligned case.

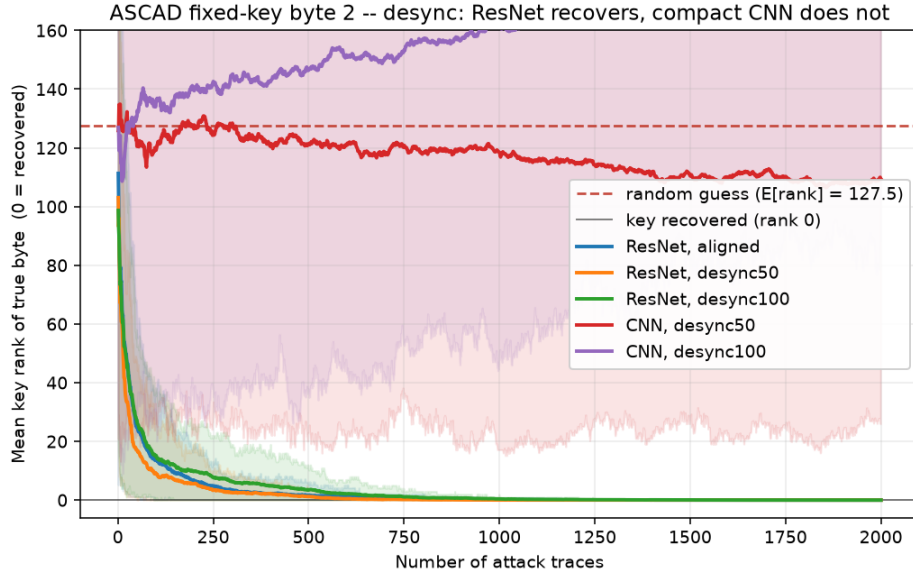


Figure 4. Fixed-key ASCAD, byte 2. The compact CNN loses the key under any misalignment; the ResNet recovers it at both shift levels.

5.6 Milestone 3d - control

As a check against recovering the key from a label-distribution artefact rather than real leakage, the CNN was retrained on the variable-key set with the profiling labels randomly permuted. Attack accuracy was 0.388%, i.e. exactly 1/256, and the key rank stayed between 90 and 128 and never reached 0. The pipeline therefore depends on genuine trace-label association.

6 Summary of results

Model	Fixed-key byte 2	Variable-key byte 2
Random forest	not recovered (rank ~63)	-
SVM (RBF, PCA-50)	not recovered (rank 138)	-
CNN (compact)	1360 traces	520 traces
ResNet (single-task)	870 traces	280 traces

ResNet (multi-task)	270 traces	3400 traces (worse)
CNN, desync 50 / 100	not recovered (rank 109 / 176)	-
ResNet, desync 50 / 100	608 / 979 traces	-
CNN, shuffled labels	-	not recovered (acc = 1/256)

Table 1. Attack traces needed to reach mean key rank 0. A dash means the configuration was not run.

7 Discussion

7.1 Deliberate deviations from the paper

Several choices differ from a literal reading of the paper, each for a stated reason. The compact CNN replaces the 60M-parameter VGG network, which does not rank the key without heavy tuning. The SVM is trained on a 10 000-trace subsample because a full RBF SVM over 50 000 traces and 256 classes is impractical. Model selection uses validation guessing entropy, not validation loss. The target is always key byte 2, since the published traces only carry its leakage. These are recorded in a table in the repository README.

7.2 What reproduced and what did not

Reproduced: the classical-versus-deep split, key recovery by both the CNN and the ResNet, the ResNet's advantage on the variable-key set, and the ResNet tolerating trace desynchronisation that defeats the compact CNN. Not reproduced: the literature's ~35-trace efficiency - our best variable-key figure is around 280 traces for a single seed, so the gap is roughly 8x and would need a proper multi-seed hyperparameter search to close. The multi-task ResNet is a clear win on the fixed-key set and a clear loss on the variable-key set, which is a useful reminder that an auxiliary objective only helps while optimisation, not capacity, is the limiting factor.

7.3 Limitations

Most numbers are single-seed, and where a training run was repeated the spread was large (variable-key ResNet: 280 vs 1540 traces; desync-50 ResNet: no recovery at 50 epochs, 608 traces at 80), so the trace counts should be read as order-of-magnitude figures and the deep models as sensitive to the epoch budget. Full-key recovery was not attempted because it needs the raw trace set. Desync was tested but not defended against with augmentation. The guessing-entropy checkpoint metric is itself noisy on small validation slices, which occasionally selects a lucky epoch.

8 Conclusion

The core attack of Poudel and Rahimi reproduces: a profiled deep model recovers a byte of a masked AES key from ASCAD traces where classical models cannot, and a 1-D ResNet does so more efficiently than a CNN on the harder dataset. A multi-task variant that also predicts the mask shares improves the fixed-key attack substantially but not the variable-key attack. The efficiency figures reported in the wider literature were not matched and remain the main target for further tuning.

References

- [1] A. Poudel and A. Rahimi. Machine Learning-Based AES Key Recovery via Side-Channel Analysis on the ASCAD Dataset. arXiv:2508.11817, 2025.
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- [4] L. Karayalcin, G. Perin and S. Picek. Resolving the Doubts: On the Construction and Use of ResNets for Side-Channel Analysis. Mathematics, 2023.
- [5] T. Marquet and E. Oswald. A Comparison of Multi-task Learning and Single-task Learning Approaches. IACR ePrint 2023/006.